

E.G.S. PILLAY ENGINEERING COLLEGE

(Autonomous)

Approved by AICTE, New Delhi | Affiliated to Anna University, Chennai

Accredited by NAAC with 'A' Grade | Accredited by

NBA(CSE, EEE, MECH, IT, ECE and CIVIL)

NAGAPATTINAM – 611002



B.TECH. INFORMATION TECHNOLOGY

R-2019

THIRD YEAR

SEMESTER V										
Course Code	Course Name	L	T	P	C	Maximum Marks			Category	
						CA	ES	Total		
Theory Course										
1902IT501	Object Oriented Analysis and Design	3	0	0	3	40	60	100	PC	
1902IT502	Web Programming	3	0	0	3	40	60	100	PC	
1902IT503	Security in Computing	3	0	0	3	40	60	100	PC	
1902IT504	Internet of Things	3	2	0	4	40	60	100	PC	
	Professional Core Elective I	3	0	0	3	40	60	100	PE	
	Humanities Science Elective I	3	0	0	3	40	60	100	HSSE	
Laboratory Course										
1902IT551	Web Programming Lab	0	0	2	1	50	50	100	PC	
1904IT551	Case Tools (Mini Project I)	0	0	2	1	50	50	100	EEC	
1904GE551	Life Skills: Aptitude I	0	0	2	1	100	-	100	EEC	
1904IT552	Startup Opportunities for IT Engineers	0	0	2	1	100	-	100	EEC	
Audit Course										
1902MCX03	Essence of Indian Traditional Knowledge	2	0	0	0	100	-	100	-	
Total		20	2	8	23	640	460	1100	-	

1902IT501		OBJECT ORIENTED ANALYSIS AND DESIGN	L	T	P	C
			3	0	0	3
AIM: To study various object oriented analysis and design method using CASE tools						
PREREQUISITE: Software Engineering and Project Management						
COURSE OBJECTIVES:						
1. Learn the basics of OO analysis and design skills 2. Learn the UML design diagrams 3. Learn to map design to code 4. Be exposed to the various testing techniques.						
UNIT I	UML DIAGRAMS					9 Hours
Introduction to OOAD – Unified Process - UML diagrams – Use Case – Class Diagrams– Interaction Diagrams – State Diagrams – Activity Diagrams – Package, component and Deployment Diagrams						
UNIT II	DESIGN PATTERNS					9 Hours
GRASP: Designing objects with responsibilities – Creator – Information expert – Low Coupling – High Cohesion – Controller - Design Patterns – creational - factory method - structural – Bridge – Adapter- behavioral – Strategy – observer						
UNIT III	CASE STUDY					9 Hours
Case study – the Next Gen POS system, Inception -Use case Modeling - Relating Use cases –include, extend and generalization - Elaboration - Domain Models - Finding conceptual classes and description classes – Associations – Attributes – Domain model refinement – Finding conceptual class Hierarchies - Aggregation and Composition						
UNIT IV	APPLYING DESIGN PATTERNS					9 Hours
System sequence diagrams - Relationship between sequence diagrams and use cases Logical architecture and UML package diagram – Logical architecture refinement - UML class diagrams – UML interaction diagrams - Applying GoF design patterns						
UNIT V	CODING AND TESTING					9 Hours
Mapping design to code – Testing: Issues in OO Testing – Class Testing – OO Integration Testing –GUI Testing – OO System Testing						
					TOTAL:	45 Hours
FURTHER READING: Software Development, Software Design						
COURSE OUTCOMES						
At the end of this course, students will able to, CO1: Design and implement projects using OO concepts CO2: Use the UML analysis and design diagrams CO3: Apply appropriate design patterns CO4: Create code from design CO5: Compare and contrast various testing techniques						
REFERENCES:						
1. Craig Larman, "Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development", Third Edition, Pearson Education, 2016.						
2. Simon Bennett, Steve Mc Robb and Ray Farmer, “Object Oriented Systems Analysis and Design Using UML”, Fourth Edition, Mc-Graw Hill Education, 2012.						
3. Erich Gamma, a n d Richard Helm, Ralph Johnson, John Vlissides, “Design patterns:Elements of Reusable Object-Oriented Software”, Addison-Wesley, 2012.						
4. Martin Fowler, “UML Distilled: A Brief Guide to the Standard Object Modeling Language”, Third edition, AddisonWesley, 2013.						
5. Paul C. Jorgensen, “Software Testing:- A Craftsman’s Approach”, Third Edition, Auerbach Publications, Taylor and Francis Group, 2010.						
6. http://nptel.ac.in/						

1902IT502		WEB PROGRAMMING	L	T	P	C
			3	0	0	3
AIM: The main objective of this course is used to introduce the concepts of Web Essentials and various web services						
PREREQUISITE: Programming in Java Programming, Database Management Systems.						
COURSE OBJECTIVES:						
1) To understand the concept of client / server programming 2) To apply web programming languages for developing web applications 3) To know the unique features of scripting languages						
UNIT I	WEB ESSENTIALS					9 Hours
Internet – Web clients – Web servers – Markup languages – Introduction to XHTML-Editing XHTML-Headings-Linking –Tables-Images-Forms-Internal linking – Frames - Lists- Cascading Style Sheets (CSS): Features-Style rule cascading and inheritance - Text properties –CSS box model.						
UNIT II	CLIENT SIDE PROGRAMMING					12 Hours
Client side vs. Server side programming languages - Introduction to java script –Control statements I - Control statements II - Functions- Objects – Arrays – PHP Programming						
UNIT III	SERVER SIDE PROGRAMMING					12 Hours
Java servlet: Architecture – Servlet life cycle -Simple programs using java servlet– Parameter data – Sessions – Cookies – Other servlet capabilities –Data storage –Servlet and concurrency- JDBC- Connecting a java servlet program to a database						
UNIT IV	XML AND WEB SERVICES					12 Hours
XML Namespaces-DTD and XML schema-XML parsers: DOM vs. SAX-XSLT – Xquery - XPath- JSP - Running JSP applications – Java beans classes and JSP - Web services concepts - Web services for clients – WSDL – Representing data types: XML schema – SOAP - J2EE						
					TOTAL:	45 Hours
FURTHER READING: Software Development, Mobile Application Development						
COURSE OUTCOMES						
At the end of this course, students will able to, CO1: Design web pages using HTML and CSS CO2: Develop web pages using java script CO3: Develop server side programming techniques to solve real time application CO4: Apply database concept to create interactive web pages CO5: Apply JSP concepts to solve real time applications CO6: Understand the basic concept of web services						
REFERENCES:						
1. Jeffrey C Jackson, Web Technology – A computer Science perspective, Person Education, New Delhi, 2016.						
2. Frank. P. Coyle, XML, Web Services and the Data Revolution, Addison-Wesley Professional, 2012.						
3. Chris Bates, Web Programming – Building Internet Applications, Wiley India, 2013.						
4. Deitel, Deitel and Neito, Internet and World wide web – How to program, Pearson education, New Delhi, 2016.						
5. Gopalan. N.P, Web Technology A Developer Perspectives, PHI, 2012						
6. H.M.Deitel, P.J.Deitel, T.R.Nieto, T.M.Lin, XML How to Program, Pearson Education,2012						
7. Ramesh Nagappan, Robert Skoczylas and Rima Patel Sriganesh, Developing Java Web Services, Wiley Publishing Inc., 2011.						
8. Steve Graham and Doug Davis, Building Web services with Java, Pearson Education 2011						
9. http://nptel.ac.in/						
10. http://sololearners.com/						
11. http://tutorialpoint.org						

1902IT503		SECURITY IN COMPUTING	L	T	P	C
			3	0	0	3
AIM: To study various cryptography techniques for enable security and study various network security protocols						
PREREQUISITES: Computer Networks, Java Programming						
COURSE OBJECTIVES:						
1. Understand the concepts of public key encryption and number theory 2. Understand authentication and hash functions. 3. Know the network security tools and applications. 4. Understand the system level security used.						
UNIT I	INTRODUCTION					9 Hours
Motivating examples– Basic concepts: confidentiality, integrity, availability, security policies, security mechanisms, assurance –Basic Cryptography: Historical background, –Elementary Ciphers (Substitution, Transposition and their Properties) –Caesar Cipher– Data Encryption Standard – Block Cipher Design Principles and Modes of Operation – Case study: AES						
UNIT II	PUBLIC KEY CRYPTOGRAPHY					9 Hours
Euclidean algorithm –Euler Theorem– Fermat Theorem– Totient functions– multiplicative and additive inverse – Selection of public and private keys–Case Study: Diffie-Hellman key Exchange – Elliptic Curve Architecture and Cryptography - Introduction to Number Theory – Confidentiality using Symmetric Encryption – Public Key Cryptography – Case Study: RSA						
UNIT III	AUTHENTICATION AND HASH FUNCTION					9 Hours
Security Handshake pitfalls–Online vs. offline password guessing–Reflection attacks Per-session keys and authentication tickets–Key distribution centers and certificate authorities Authentication requirements – Authentication functions – Message Authentication Codes– Hash Functions – Case Study: MD5, HMAC.						
UNIT IV	NETWORK SECURITY AND FIREWALLS					9 Hours
Public Key infrastructures– IPSec - IKE– SSL/TLS - Authentication Application: X.509 Authentication Service – Electronic Mail Security – PGP – S/MIME - IP Security – Web Security – Kerberos - Packet filters– Application level gateways– Encrypted tunnels						
UNIT V	HACKING					9 Hours
Introduction to Hacking – Hacking Process – Foot printing – System Hacking – Trojan Horses – Ethical Hacking – Attacks and Countermeasures						
TOTAL:						45 Hours
FURTHER READING		Cyber Forensics, Security Management issues				
COURSE OUTCOMES						
After learning the course the student should be able to: CO1: Explain concepts related to applied cryptography, including symmetric cryptography, asymmetric cryptography, and digital signatures CO2: Understand the theory behind the security of different cryptographic algorithms. CO3: Understand common network vulnerabilities, defense mechanisms against network attacks, and cryptographic protection mechanisms. CO4: Apply the requirements of non-realtime security (email security) and ways to provide privacy, source authentication, message integrity, non-repudiation, proof of submission, proof of delivery, message flow confidentiality, and anonymity.						
REFERENCES:						
1. William Stallings, “Cryptography and Network Security – Principles and Practices”, Pearson Education, Third Edition, 2016 2. Charlie Kaufman, Radia Perlman, and Mike Speciner, “ Network Security: PRIVATE Communication in a PUBLIC World”, Prentice Hall, ISBN 0-13-046019-2, 2017 3. Behrouz A. Foruzan, “Cryptography and Network Security”, Tata McGraw-Hill, 2013 4. Bruce Schneier, “Applied Cryptography”, John Wiley & Sons Inc, 2013. 5. Charles B. Pfleeger, Shari Lawrence Pfleeger, “Security in Computing”, Third Edition, Pearson Education, 2012 6. Wade Trappe and Lawrence C. Washington , “ Introduction to Cryptography with coding theory” , Pearson Education, 2012. 7. Thomas Calabrese, “Information Security Intelligence : Cryptographic Principles and Applications”, Thomson Delmar Learning, 2012. 8. http://nptel.ac.in/						

1902IT504		INTERNET OF THINGS	L	T	P	C
			3	2	0	4
AIM:The main objective of this course is to understand the IoT is presently a hot technology worldwide. Government, academia, and industry are involved in different aspects of research, implementation, and business with IoT.						
PREREQUISITE		Basic Programming Knowledge, Computer Architecture				
COURSE OBJECTIVES:						
1. Study the concept of Microprocessor and Microcontrollers						
2. Study what is Internet of Thing and learning concepts						
3. Get basic knowledge of RFID technology, sensor technology and satellite technology						
4. Students aware of resource management and security issues in Internet of Things						
5. Study the concept of Internet of Things in the real world scenario						
UNIT I	INTRODUCTION TO IOT					12 Hours
What is the Internet of Things? : History of IoT, About IoT, Overview and Motivations, Examples of Applications, Internet of Things Definitions and Frameworks: IoT Definitions, IoT Architecture, General Observations, ITU-T Views, Working Definition, IoT Frameworks, Basic Nodal Capabilities						
UNIT II	IOT PROTOCOLS					12 Hours
Sensors - Protocol Standardization for IoT – Efforts – M2M and WSN Protocols – SCADA and RFID Protocols – Issues with IoT Standardization – Unified Data Standards						
UNIT II	CASE STUDY					12 Hours
IEEE 802.15.4 – BACNet Protocol– Modbus – KNX – Zigbee Architecture - Software & Management Tools for IoT						
UNIT IV	BUILDING IOT WITH RASPBERRY PI AND GALILEO/ARDUINO					12 Hours
Physical device – Raspberry Pi Interfaces – Programming – APIs / Packages – Web services - Intel Galileo Gen2 with Arduino- Interfaces - Arduino IDE – Programming - APIs and Hacks – path planning – obstacle avoidance technique						
UNIT V	EXPERIMENTS USING IOT					12 Hours
Home Automation – Cities – Environment – Energy – Retail – Logistics - Agriculture - Industry - Health and Lifestyle - IoT and M2M						
REQUIREMENTS: Raspberry PI and Arduino Tool Kit						
TOTAL:						60 Hours
COURSE OUTCOMES						
At the end of this course, students will able to,						
CO1: Apply microprocessor and microcontrollers concepts to solve various IoT problems						
CO2: Explain the concepts of IoT and protocols						
CO3: Illustrate various case studies and protocol architecture						
CO4: Develop a portable IOT using Arduino or equivalent boards and relevant protocols.						
CO5: Analyze applications of IOT in real time scenario.						
REFERENCES:						
1. Romesh Gaonkar , “Microprocessor Architecture, Programming, and Applications with the 8085”, Penram International Publishing (India) LTD, 2017						
2. Yu-Cheng Liu, Glenn A.Gibson, “Microcomputer Systems: The 8086 / 8088 Family - Architecture, Programming and Design”, Second Edition, Prentice Hall of India, 2014.						
3. Mohamed Ali Mazidi, Janice Gillispie Mazidi, Rolin McKinlay, “The 8051 Microcontroller and Embedded Systems: Using Assembly and C”, Second Edition, Pearson Education, 2013.						
4. Daniel Minoli, “Building the Internet of Things with IPv6 and MIPv6: The Evolving World of M2M Communications”, ISBN: 978-1-118-47347-4, Willy Publications. 2017						
5. Bernd Scholz-Reiter, Florian Michahelles, “Architecting the Internet of Things”, ISBN 978-3- 642-19156-5 e-ISBN 978-3-642-19157-2, Springer.,2016						
6. Arshdeep Bahga, Vijay Madiseti, “Internet of Things – A hands-on approach”, Universities Press, 2015						
7. http://nptel.ac.in/ & http://coursera.org/						

1902IT551		WEB PROGRAMMING LAB	L	T	P	C
			0	0	2	1
PREREQUISITE: Programming in Java Programming, Database Management Systems.						
LIST OF EXPERIMENTS						
1. Write a html program for Creation of web site with forms, frames, links, tables etc 2. Design a web site using HTML and DHTML. Use Basic text Formatting, Images, 3. Create a script that asks the user for a name, then greets the user with "Hello" and the user name on the page 4. Create a script that collects numbers from a page and then adds them up and prints them to a blank field on the page. 5. Create a script that prompts the user for a number and then counts from 1 to that number displaying only the odd numbers. 6. Create a script that will check the field in Assignment 1 for data and alert the user if it is blank. This script should run from a button. 7. Using CSS for creating web sites 8. Creating simple application to access data base using JDBC Formatting HTML with CSS. 9. Program for manipulating Databases and SQL. 10. Program using PHP database functions. 11. Write a web application that functions as a simple hand calculator, but also keeps a "paper trail" of all your previous work 12. Install Tomcat and use JSP and link it with any of the assignments above 13. Reading and Writing the files using .Net 14. Write a program to implement web service for calculator application 15. Implement RMI concept for building any remote method of your choice.						
Requirements: Web Browser, JDK1.5, Netbeans IDE or Eclipse or Equivalent						
TOTAL:						30 Hours

1904IT551		CASE TOOLS (MINI PROJECT 1)	L	T	P	C
			0	0	2	1
AIM: The objective of this course is student develop mini project using CASE tools						
PREREQUISITE: Software Engineering and Project Management						
COURSE OBJECTIVES:						
1. Learn the basics of OO analysis and design skills. 2. Be exposed to the UML design diagrams. 3. Learn to map design to code. 4. Be familiar with the various testing techniques						
TO DEVELOP A MINI-PROJECT USING FOLLOWING PROBLEM STATEMENTS						
✓ Identify Use Cases and develop the Use Case model. ✓ Identify the conceptual classes and develop a domain model with UML Class diagram. ✓ Using the identified scenarios, find the interaction between objects and represent them using UML Sequence diagrams. ✓ Draw relevant state charts and activity diagrams. ✓ Identify the User Interface, Domain objects, and Technical services. Draw the partial layered, logical architecture diagram with UML package diagram notation. ✓ Develop and test the Technical services layer. ✓ Develop and test the Domain objects layer. ✓ Develop and test the User interface layer.						
Suggested domains for Mini-Project (not Limited too):						
• Passport automation system • Book bank • Online Examination • Online course reservation system • E-ticketing • E-Commerce • Recruitment system • Foreign trading system • Conference Management System • BPO Management System • Library Management System						

Requirements: Argo UML or Eclipse IDE or Rational Suite or Visual Paradigm or equivalent	
	TOTAL: 45 Hours
COURSE OUTCOMES	
At the end of this course, students will able to, CO1: Design and implement projects using OO concepts. CO2: Use the UML analysis and design diagrams. CO3: Apply appropriate design patterns. CO4: Create code from design. CO5: Compare and contrast various testing techniques	

1904IT552		STARTUP OPPORTUNITIES FOR IT ENGINEERS	L	T	P	C
			0	0	2	1
AIM: This course helps you understand the process of entrepreneurship from a technology-oriented background						
COURSE OBJECTIVES:						
	1. Learn what it takes to become a “technopreneur” 2. Explore various methods for identifying opportunities 3. Learn how to conduct market research and provide evidence for the viability of the business idea 4. Develop a viable business proposition and learn to pitch your ideas for various audiences 5. Understand the dynamics of new venture development and team building 6. Develop the ability to translate a business idea into marketing and financial plans					
Course Contents						
	1. Introduction to Startups 2. Innovation & Entrepreneurship 3. Entrepreneurial Mindset, Entrepreneurial Skillset 4. Global Startup Eco System 5. IT based Startup Trends 6. Problems – Identification, Selection & Validation 7. Evolution of Technology & Startups 8. Startup Opportunity Identification 9. Business Model Trends – Examples 10. Startup Idea & Technology Landscape 11. Digital Marketing 12. Minimum Viable Product Development & Tools for Creating MVPs 13. Patentability 14. Raising Capital & Funding Models 15. Legal Procedures & Launchpad					
			TOTAL:		30 HOURS	
COURSE OUTCOMES:						
	After completion of the course, Student will be able to					
CO1	Explain Concepts of Innovation, Entrepreneurship and Startups in Technology					
CO2	Develop Startup or Business ideas, minimum viable products and business models for real life problems					
Evaluation Procedure						
	1. Assignment 1– Review of IT based Startups – 20 Marks 2. Assignment 2– Problems, Problem Identification, Selection & Evaluation – 30 Marks 3. Assignment 3 – Review of Business Models of IT based Startups – 20 Marks 4. Presentation – Idea, Market Research Results, 1st MVP, Possible Fund Raising Model to be adopted, IP – 30 Marks					
REFERENCES:						
1. The High-Performance Entrepreneur by Subroto Bagchi						
2. The Law of Success in Sixteen Lessons Paperback by Napoleon Hill						
3. The E-Myth Revisited: Why Most Small Businesses Don’t Work and What to Do About It by Michael E. Gerber						
4. https://www.edx.org/course/entrepreneurship-for-engineers						

1704GE551	LIFE SKILLS: APTITUDE – 1	L	T	P	C
		0	0	2	1
Course Objectives: The students should be made to: 1.To brush up problem solving skill and to improve intellectual skill of the students 2.To be able to critically evaluate various real life situations by resorting to Analysis Of key issues and factors 3.To be able to demonstrate various principles involved in solving mathematical problems and thereby reducing the time taken for performing job functions. 4.To enhance analytical ability of students 5.To augment logical and critical thinking of Student					
Unit I	INTRODUCTION TO NUMBER SYSTEM, BASIC SHORTCUTS OF ADDITION, MULTIPLICATION, DIVISION	6 Hours			
Classification of numbers – Types of Numbers - Divisibility rules - Finding the units digit - Finding remainders in divisions involving higher powers - LCM and HCF Models - Fractions and Digits – Square, Square roots – Cube, Cube roots – Shortcuts of addition, multiplication, Division.					
Unit II	RATIO AND PROPORTION, AVERAGES	6 Hours			
Definition of Ratio - Properties of Ratios - Comparison of Ratios - Problems on Ratios - Compound Ratio - Problems on Proportion, Mean proportional and Continued Proportion Definition of Average - Rules of Average - Problems on Average - Problems on Weighted Average - Finding average using assumed mean method.					
Unit III	PERCENTAGES, PROFIT AND LOSS	6 Hours			
Introduction Percentage - Converting a percentage into decimals - Converting a Decimal into a percentage - Percentage equivalent of fractions - Problems on percentages - Problems on Profit and Loss percentage- Relation between Cost Price and Selling price - Discount and Marked Price - Two different articles sold at same Cost Price - Two different articles sold at same Selling Price - Gain% / Loss% on Selling Price.					
Unit IV	CODING AND DECODING, DIRECTION SENSE	6 Hours			
Coding using same set of letters - Coding using different set of letters - Coding into a number - Problems on R-model - Solving problems by drawing the paths - Finding the net distance travelled - Finding the direction - Problems on clocks - Problems on shadows - Problems on direction sense using symbols and notations.					
Unit V	NUMBER AND LETTER SERIES NUMBER AND LETTER ANALOGIES, ODD MAN OUT	6 Hours			
Difference series - Product series - Squares series - Cubes series - Alternate series - Combination series - Miscellaneous series - Place values of letters - Definition of Analogy - Problems on number analogy - Problems on letter analogy - Problems on verbal analogy - Problems on number Odd man out - Problems on letter Odd man out - Problems on verbal Odd man out					
TOTAL					30 Hours
COURSE OUTCOMES: At the end of the course, the student should be able to CO1: Learners should be able to understand number and solving problems least time using various shortcut CO2: Solve problems on averages; compare two quantities using ratio and proportion. CO3: Calculate concept of percentages, implement business transactions using profit and loss. CO4: Workout concepts of Coding and Decoding, ability to visualize directions and understand the logic behind a sequence. CO5: Learners should be able to find a series the logic behind a sequence.					
REFERENCES: 1. Arun Sharma, ‘How to Prepare for Quantitative Aptitude for the CAT’, 7 th edition, McGraw Hills publication, 2016. 2. Arun Sharma, ‘How to Prepare for Logical Reasoning for CAT’, 4 th edition, McGraw Hills publication, 2017. 3. R S Agarwal, ‘A modern approach to Logical reasoning’, revised edition, S.Chand publication, 2017. 4. R S Agarwal, ‘Quantitative Aptitude for Competitive Examinations’, revised edition, S.Chand publication, 2017. 5. Rajesh Verma, “Fast Track Objective Arithmetic”, 3 rd edition, Arihant publication, 2018. 6. B.S. Sijwalii and Indu Sijwali, “A New Approach to REASONING Verbal & Non-Verbal”, 2 nd edition, Arihant publication, 2014.					
ASSESSMENT PATTERN : 1. Two tests will be conducted (25 * 2) - 50 marks 2. Five assignments will be conducted (5*10) - 50 Marks.					

1902MCX03	ESSENCE OF INDIAN TRADITIONAL KNOWLEDGE		L	T	P	C
			2	0	0	0
COURSE OBJECTIES:						
	1. The course aims at imparting basic principles of thought process, reasoning and inferencing. Sustainability is at the core of Indian Traditional Knowledge Systems connecting society and nature. 2. Holistic life style of Yogic-science and wisdom capsules in Sanskrit literature are also important in modern society with rapid technological advancements and societal disruptions. 3. The course focuses on introduction to Indian Knowledge System, Indian perspective of modern scientific world-view and basic principles of Yoga and holistic health care system.					
1. Basic Structure of Indian Knowledge System (i) Vedas, (ii) Uveda (Ayurveda, Dhanurveda, Gandhaveda, Sthaitya Adad) (iii) Vedang (Shiksha, Kalla, Nanrut, Grammar, Jyotisha Chhanda), (iv) Uraiga (Dharma Vastra, Shringa, Guarana, Tirmasra) 2. Modern Science and Indian Knowledge System 3. Yoga and Holistic Health care 4. Case Studies.						
			TOTAL:	30 HOURS		
COURSE OUTCOMES:						
After completion of the course, Student will be able to understand , connect up and explain basics of Indian Traditional knowledgemodern scientific perspective						
REFERENCES:						
1. V. Sivaramakrishna (Ed.), Cultural Heritage of India-Course Material, Bharatiya Vidya Bhavan, Mumbai,5th Edition, 2014 2. Swami Jitatmanand, Modern Physics and Vedant, Bharatiya Vidya Bhavan, Fritzof Capra, Tao of Physics 3. Fritzof Capra, The wave of Life 4. V N Jha (Eng. Trans,), Tarkasangraha of Annam Bhatta, Inernational Chinmay Foundation, Velliarnad,Amaku,am 5. Yoga Sutra of Patanjali, Ramakrishna Mission, Kolkatta 6. GN Jha (Eng. Trans.) Ed. R N Jha, Yoga-darshanam with Vyasa Bhashya, Vidyanidhi Prakasham, Delhi,2016 7. RN Jha, Science of Consciousness Psychotherapy and Yoga Practices, Vidyanidhi Prakasham, Delhi, 2016 8. P R Sharma (English translation), Shodashang Hridayam						